*) Chapter #13 (Numericals) (Pg #303)

Question #01:-

Data:

$\lambda = 6900\,°A = 6900 \times 10^{-10}\,m$

Distance of screen $(L) = 3.3\,m$

Fringe width $(x) = 1.80\,cm = 0.018\,m$

Distance b/w the slits $(d) = ?$

Solution:

$$x = \frac{\lambda L}{d}$$

$$d = \frac{\lambda L}{x}$$

$$= \frac{(6900 \times 10^{-10}) \times (3.3)}{0.018}$$

$$[\, d = 1.265 \times 10^{-4}\,m \,]$$

Question #02:-

Data:

Distance of mirror $(x) = 25.8\,\mu m = 25.8 \times 10^{-6} m$
No. of fringes $(m) = 92$
$\lambda = ?$

Solution:

For Michelson Interferometer

$$x = \frac{m\lambda}{2}$$

$$\lambda = \frac{2x}{m}$$

$$= \frac{(2)(25.8 \times 10^{-6})}{92}$$

$$[\lambda = 5.6 \times 10^{-7} m]$$

$$= 560 \times 10^{-9} m$$

OR

$$[\lambda = 560 nm]$$

Question #04:-

Data:

$\lambda = 400\ nm = 400 \times 10^{-9} m$

thickness $(\Delta t) = t_6 - t_3 = ?$

$R = 5m$

$r_3 = ?$

Solution:

For bright fringe in Newton's Ring

$$r_n = \sqrt{\left(N - \frac{1}{2}\right)\lambda R}$$

$$r_3 = \sqrt{\left(3 - \frac{1}{2}\right)(400 \times 10^{-9})(5)}$$

$$\left[r_3 = 2.236 \times 10^6\right]$$

OR

$$\left[r_3 = 2.236mm\right]$$

For bright fringe in thin film

$$2t = \left(m + \frac{1}{2}\right)\lambda$$

For 3rd bright fringe, $m = 2$

$$2t_3 = \left(2 + \frac{1}{2}\right) 400 \times 10^{-9}$$

$$2t_3 = \frac{5}{2} \times 400 \times 10^{-9}$$

$$[t_3 = 5 \times 10^{-7} m]$$

For 6th bright fringe, $m = 5$

$$2t_6 = \left(5 + \frac{1}{2}\right) 400 \times 10^{-9}$$

$$2t_6 = \frac{7}{2} \times 400 \times 10^{-9}$$

$$[t_6 = 1.1 \times 10^{-6} m]$$

For Δt:-

$$\Delta t = t_6 - t_3$$
$$= 1.1 \times 10^{-6} - 5 \times 10^{-7}$$
$$= 6 \times 10^{-7}$$

$f = 400 - 700$ (Range)
OR
$380 - 750$

$$= 600 \times 10^{-9} m$$
$$[\Delta t = 600 nm] \text{ Ans}$$

Question #05:-

Data:

$n = 1.5$
$t = 910 nm$
a) Missing wavelength = ?
b) Strongest wavelength = ?

Solution:

a) For missing wavelength using the formula for destructive interference in the thin film.

$$2nt = m\lambda$$
$$2(1.5)(910) = (1)\lambda$$
$$[\lambda_1 = 2730 nm]$$

For $m=2$,

$$2nt = m\lambda$$

$$\lambda = \frac{2nt}{m}$$

$$\lambda_2 = \frac{2730}{2}$$

$$\left[\lambda_2 = 1365\,nm\right]$$

For $m=3$

$$\lambda_3 = \frac{2730}{3}$$

$$\left[\lambda_3 = 910\,nm\right]$$

For $m=4$

$$\lambda_4 = \frac{2730}{4}$$

$$\left[\lambda_4 = 682.5\,nm\right]$$

For $m=5$

$$\lambda_5 = \frac{2730}{5}$$

$$\left[\lambda_5 = 546\,nm\right]$$

For $m=6$;

$$\lambda_6 = \frac{2730}{6}$$

$$\left[\lambda_6 = 455\,nm\right]$$

For $m=7$;

$$\lambda_7 = \frac{2730}{7}$$

$$\left[\lambda_7 = 390\,nm\right]$$

b) For strongest wavelength in their films, use formula for constructive interference.

For $m=0$;

$$2nt = \left(m + \frac{1}{2}\right)\lambda$$

$$2(1.5)(910) = \left(0 + \frac{1}{2}\right)\lambda$$

$$\left[\lambda = 5460\,nm\right]$$

For $m=1$;

$$2(1.3)(910) = \left(1+\frac{1}{2}\right)\lambda$$

$$2730 = \frac{3}{2}\lambda$$

$$\lambda_2 = \frac{2730 \times 2}{3} = 1820\,nm$$

For $m=2$;

$$\lambda_3 = \frac{2730 \times 2}{5} = 1092\,nm$$

For $m=3$;

$$\lambda_4 = \frac{2730 \times 2}{7} = 780\,nm$$

For $m=4$;

$$\lambda_5 = \frac{2730 \times 2}{9} = 606.6\,nm$$

For $m=5$;

$$\lambda_6 = \frac{2730 \times 2}{11} = 486.36\,nm$$

For $m=6$;

$$\lambda_7 = \frac{2730 \times 2}{13}\ 420\,nm$$

Since visible range of reflected light is $400 \rightarrow 700$. so missing wavelengths are 2730 nm, 1365nm, 910nm

$\Rightarrow$ Strongest wavelength is $\lambda_5 = 606.6\,nm$

Question #12:-

Data:

width $(d) = 0.02mm = 20 \times 10^{-3} m$
distance of screen $(L) = 1.2m$
$\lambda = 430\,nm = 430 \times 10^{-9} m$
Width of central maximum $(\Delta x) = ?$

Solution:

$$\Delta x = \frac{2\lambda L}{d}$$

$$\Delta x = \frac{2 \times 430 \times 10^{-9} \times 1.2}{0.02 \times 10^{-3}}$$

$$[\Delta x = 0.0516\, m]$$

OR

$$[\Delta x = 5.16\, cm]$$

Question # 13:-

Data:

No. of lines $(N) = 8000$
Length of grating $(L) = 2.54\ cm = 2.54 \cdot 10^{-2} m$
$\lambda = 546nm = 546 \times 10^{-9} m$
$m = 3$
$\theta_3 = ?$

Solution:

For Grating:

$$d\sin\theta = m\lambda \cdots (i)$$

First find 'd':-

$$d = \frac{L}{N} = \frac{2.54 \times 10^{-2}}{8000}$$

$$[d = 3.175 \times 10^{-6} m]$$

Put the value of 'd' in eq (i):-

$$d\sin\theta = m\lambda$$

$$\sin\theta_3 = \frac{m\lambda}{d}$$

$$\sin\theta_3 = \frac{(3)(546 \times 10^{-9})}{3.175 \times 10^{-6}}$$

$$\sin\theta_3 = 0.5159$$

$$\theta_3 = \sin^{-1}(0.5159)$$
$$[\theta_3 = 31^\circ] \text{ Ans}$$

Question #14:-

Data:

$m = 1$
$\theta = 30^\circ$
$\lambda = 6 \times 10^{-5}\ cm$
No. of lines/cm = ?

Solution:

For Grating:-

$$d\sin\theta = m\lambda \cdots (i)$$
$$d\sin(30) = 1 \times 6 \times 10^{-5}$$
$$d = \frac{6 \times 10^{-5}}{\sin(30)}$$
$$[d = 1.2 \times 10^{-4}\ cm]$$

$$d = \frac{L}{N}$$
$$N = \frac{L}{d}$$

$$N = \frac{1}{1.2 \times 10^{-4}}$$

$$N = 8333$$

[No. of lines /cm = 8333] Ans

Question #15:-

Data:

$\lambda = 450 nm = 450 \times 10^{-9} m$

No. of lines = 5000/cm

No. of orders = ?

Angles = ?

Solution:

For Grating:

$$d \sin\theta = m\lambda \cdots (i)$$

$$d = \frac{L}{N}$$

$$= \frac{10^{-2}}{5000}$$

$$[d = 2 \times 10^{-6} m]$$

$\because d\sin\theta = m\lambda$

$$\sin\theta = \frac{m\lambda}{d}$$

For $m=1$

$$\sin\theta_1 = \frac{1\times 450\times 10^{-9}}{2\times 10^{-6}}$$

$$\theta_1 = \sin^{-1}(0.225)$$

$$[\theta_1 = 13°]$$

For $m=2$

$$\sin\theta_2 = \frac{2\times 450\times 10^{-9}}{2\times 10^{-6}}$$

$$\theta_2 = \sin^{-1}(0.45)$$

$$[\theta_2 = 26.7°]$$

For $m=3$

$$\sin\theta_3 = \frac{3\times 450\times 10^{-9}}{2\times 10^{-6}}$$

$$\theta_3 = \sin^{-1}(0.675)$$

$$[\theta_3 = 42.4°]$$

For $m=4$

$$\sin\theta_4 = \frac{4\times 450\times 10^{-9}}{2\times 10^{-6}}$$

$$\theta_4 = \sin^{-1}(0.9)$$

$$[\theta_4 = 64.1°]$$

For $m=5$

$$\sin\theta_5 = \frac{5\times 450\times10^{-9}}{2\times10^{-6}}$$

$$\theta_5 = \sin^{-1}(1.125)$$ [Not possible]

Answer:

order $=4$

$\theta_1 = 13°$

$\theta_2 = 26.7°$

$\theta_3 = 42.4°$

$\theta_4 = 64.1°$

Question #17:-

Data:

$\lambda = 0.071\,nm = 0.071\times10^{-9}\,m$

$d = 1.98\,Å = 1.98\times10^{-10}\,m$

$m = 2$

$\theta = ?$

Solution:

According to Bragg's Law:

$$2d\sin\theta = m\lambda$$

$$\sin\theta = \frac{m\lambda}{2d}$$

$$\sin\theta = \frac{2 \times 0.071 \times 10^{-9}}{2(1.98 \times 10^{-10})}$$

$$\sin\theta = 0.3535$$

$$\theta = \sin^{-1}(0.3535)$$

$$[\theta = 20.7^\circ]$$

OR

$$[\theta = 21^\circ]$$